

Signal Encoder

Patch Embedding



Conv1D

GeLU

Conv1D



hidden representations z_i

Positional Embedding



hidden representations z_i



learnable position embeddings p_i



physical-time embeddings t_i



BiMamba Backbone

Single BiMamba Block

RMSNorm

Forward
Mamba

Backward
Mamba

Concatenate

Linear

SiLU

Dropout



K

Representation MLP

Linear

GeLU

Linear

Dropout

RMSNorm



signal representations R

Auxiliary Classifier

Mean Pooling

Linear

GeLU

Linear

Stage1 $\mathcal{L}_{cls}$

Reconstruction Decoder

Linear

GeLU

Dropout

Linear

Reshape&Overlap-add

Stage1 $\mathcal{L}_{rec}$